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Article

Outcome-based Education: An Outline

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Abstract

Things we can do because of learning are called outcomes of learning. Outcome based education (OBE) was propounded by William Spady in the 90s to bring the focus of formal education to what the students learn rather than what they were taught. OBE is a system of education giving priority to ends, purpose, accomplishments, and results. All decisions about the curriculum, assessment, and instruction are driven by the exit learning outcomes the students should display at the end of a program or a course. This paper presents a method of writing outcomes for General higher education programs. Outcomes for a higher education program are defined at three levels as program outcomes (POs), program specific outcomes (PSOs), and course outcomes (COs). The most important aspect of an outcome is that it should be observable and measurable. These are best written in a well-defined framework of taxonomy of learning. Bloom's taxonomy of learning identifies three domains of learning: Cognitive, affective and psychomotor. Revised Bloom taxonomy of cognitive domain has two dimensions cognitive levels and knowledge categories. It is proposed that CO statements be written within a well-defined structure: Action, knowledge elements, conditions, and criteria. Tagging COs with POs, PSOs, cognitive levels and the number of classroom hours associated facilitates the computation of attainment of COs, POs, and PSOs.

Keywords

Outcome based education, program outcomes, program specific outcomes, course outcomes, revised Bloom taxonomy, action verbs, knowledge categories, attainment of outcomes

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Introduction

Learning is supposed to have occurred when we can do something that we were not able to do earlier. Learning leads to acquiring new knowledge, behaviours, skills, values, preferences or understanding and sometimes involves synthesizing different types of information (Kolb, 1984). It is the process whereby knowledge is created through the transformation of experience. Outcomes of learning state explicitly what the student is able to do at the end of a learning experience. Learning experiences in formal higher education programmes can be identified as learning units. A unit of learning maybe a few hours of self/classroom learning activity, a one-semester course or a formal programme of two to four years duration.

Outcome-based Education

It was in the early 1990s, William G. Spady propounded Outcome-Based Education (OBE) as a means to ensure quality in the American school system. Eventually, OBE was extended to the higher education system as well. According to the propounder of the system, 'OBE means focusing and organizing an institute's entire programmes and instructional efforts around the clearly defined outcomes we want all students to demonstrate when they leave the institute'(Spady, 1994). OBE is a system of education giving priority to ends, purposes, accomplishments and results. It is a realistic approach adopted worldwide today as part of the strategy of quality assurance, in which decisions about the curriculum and instruction are driven by the exit learning outcomes that the students should display at the end of a programme or a course.

Learning Outcomes are also referred to as Intended Learning Outcomes, Instructional Objectives, Educational Objectives, Behavioural Objectives, Performance Objectives, Terminal Objectives, Subordinate Skills, Subordinate Objectives, General Instructional Objectives, Specific Learning Outcomes and Competencies. An educational outcome is what the student should be able to do after the successful completion of an academic programme/course/instructional unit. Some of the main advantages of OBE are the explicitness in its relevance, potential of discourse, inherent clarity, accountability, self-directedness, flexibility, and an integrated framework of teaching and learning, and assessment (Davis and Winch, 2015). OBE accommodates different learning styles and enables innovation in teaching.

Outcomes

This note presents a method of writing outcomes for General (Sciences, Social Sciences, Humanities, and Arts) undergraduate (three-year and four-year) degree programmes. The proposed method is in the framework of current pedagogical theories and was heavily field tested through faculty development workshops. Material from the indicated references was used liberally in preparing this document.

Outcomes are the abilities the students acquire at the end of a learning experience. The learning experience can be an instructional unit that involves a small number of hours of instructional activity, a course of one-semester duration, or a two- to four-year formal undergraduate programme. Outcomes serve as the basis for effective interaction among concerned stakeholders. The outcome being the product of learning, it may be called a learning product. Therefore, 'product defines the process' in OBE. It is the results-oriented thinking and is the opposite of input-based education, where the emphasis on the educational process and where we are happy to accept whatever is the result.

Outcomes can be defined at three different levels in the case of general undergraduate programmes: (a) Programme Outcomes (POs) or statements that describe what the students graduating from general programmes should be able to do, (b) Programme Specific Outcomes (PSOs) which are statements that describe what the graduates of a specific programme should be able to do and (c) Course Outcomes (COs) or statements that describe what the students should be able to do at the end of a course.

A general undergraduate degree is a terminal degree for the majority of graduates (>80%). They get into employment not necessarily related directly to the discipline of the programme they graduated from. Whatever be the profession the graduates get into, they need to have some abilities and attitudes that make them good employees and contribute to wealth generation and service activities of the organization they work for. In any organization, the employees have to work as teams arranged in some hierarchy, communicate well in verbal and written form with peers and customers, understand the impact of what they do on the society and so on. POs represent the knowledge, skills and attitudes; all students are required to attain at the time of graduation from any programme. POs need to be identified by the university/institute offering general programmes. POs are non-specific to the discipline of the programme. Sometimes, these are referred to as liberal education and common core competencies.

A set of POs identified in the spirit of POs of institutions and organizations (for example AACU) across the world are as follows:

PO1. **Critical Thinking**: Take informed actions after identifying the assumptions that frame our thinking and actions, checking out the degree to which these assumptions are accurate and valid and looking at our ideas and decisions (intellectual, organizational and personal) from different perspectives

PO2. **Problem Solving**: Understand and solve problems of relevance to society to meet the specified needs using the knowledge, skills and attitudes acquired from humanities/sciences/mathematics/social sciences

PO3. **Computational Thinking**: Understand data-based reasoning through the translation of data into abstract concepts using computing technology-based tools

PO4. **Effective Communication**: Speak, read, write and listen clearly in person and through electronic media in English and one Indian language and make meaning of the world by connecting people, ideas, books, media and technology

PO5. **Social Interaction**: Elicit views of others, mediate disagreements and help reach conclusions in group settings
PO6. **Self-directed and Life-long Learning**: Acquire the ability to engage in independent and life-long learning in the broadest context socio-technological changes
PO7. **Effective Citizenship**: Acquire empathetic social concern and equity-centred national development and the ability to act with an informed awareness of issues and participate in civic life through volunteering
PO8. **Ethics**: Recognize different value systems, including your own, understand the moral dimensions of your decisions and accept responsibility for them
PO9. **Environment and Sustainability**: Understand the impact of technology and business practices in societal and environmental contexts and sustainable development
PO10. **Global Perspective**: Understand the economic, social and ecological connections that link the world's nations and people

The PSOs are outcomes that are specific to a programme. They characterize the specificity of the core courses of a programme. The PSOs of a general programme can only be two to four in number. For example, the sample the PSOs for BSc (Zoology) can be as follows:

PSO1. Understand the nature and basic concepts of Cell biology, Biochemistry, Taxonomy and Ecology
PSO2. Analyse the relationships among animals, plants and microbes
PSO3. Perform procedures as per laboratory standards in the areas of Biochemistry, Bioinformatics, Taxonomy, Economic Zoology and Ecology
PSO4. Understand the applications of biological sciences in Apiculture, Aquaculture, Agriculture, and Medicine

The COs represent what the students should be able to do at the end of a course. Some sample the COs from different courses are as follows:

1. Understand aspects of human development including pregnancy, parturition, birth control, infertility, developmental defects and miscarriage
2. Synthesize specified chemicals and characterize them and interpret spectral data to elucidate the structure of the synthesized chemical compound
3. Write programmes for one-dimensional and two-dimensional array manipulation and string-handling functions

Taxonomy of Learning

In 1956, Benjamin Bloom headed a group of educational psychologists who developed a classification of levels of intellectual behaviour important in learning. This became a taxonomy, including three overlapping domains: the cognitive,

psychomotor and affective. Cognitive learning is demonstrated by knowledge recall and the intellectual skills: comprehending information, organizing ideas, analysing and synthesizing data, applying knowledge, choosing among alternatives in problem-solving and evaluating ideas or actions. This domain on the acquisition and use of knowledge is predominant in majority of courses. Bloom identified six levels within the cognitive domain, from the simple recall or recognition of facts, as the lowest level, through increasingly complex and abstract mental levels, to the highest order, which is classified as evaluation. The six progressive stages of cognitive thinking are identified as *knowledge (recall), comprehension, application, analysis, synthesis and evaluation*.

Krathwohl (1964) took the lead to produce a parallel taxonomy explaining the development of attitudes, principles, codes and human values. Affective learning is demonstrated by behaviours indicating attitudes of awareness, interest, attention, concern and responsibility, ability to listen and respond in interactions with others and ability to demonstrate those attitudinal characteristics or values which are appropriate to the test situation and the field of study.

Kibler, Barker and Miles (1970) completed the trilogy of taxonomies with the physical dimensions of behaviour, as it develops from gross to fine movements and nonverbal to verbal activities. Psychomotor learning is demonstrated by physical skills; coordination, dexterity, manipulation, grace, strength, speed; actions which demonstrate the fine motor skills, such as the use of precision instruments or tools, or actions which evidence gross motor skills, such as the use of the body in dance or athletic performance.

Together, these taxonomies for cognitive learning, social interaction and physical development are the recognized building blocks for creating measurable learning outcomes, planning instruction and measuring the attainment of outcomes.

Bloom's taxonomy has been revisited several times by educational psychologists, and several variants of the original taxonomy were proposed. L. W. Anderson, D. R. Krathwohl and others presented a revision of Bloom's taxonomy of educational objectives in 2001 to re-establish the relevance of the ideas in Handbook (1956) and to incorporate new knowledge and thought that had been produced since 1956. The revised framework was intended to broaden the typical set of learning outcomes that promote 'retention' and 'transfer'. There exist several other taxonomies too (Biggs & Collis 1982; Fink, 2013; Gagné, 1985; Gagné & Biggs, 1974; Marzano & Kendall, 2001). All taxonomies are attempts to give a structure to the processes involved in learning based on the observations of learning behaviours and the limited understanding of how the brain functions. Nevertheless, most academics focus on Anderson–Bloom Taxonomy, also referred to as Revised Bloom's Taxonomy (Anderson et al., 2001).

Cognitive Levels

Cognitive learning is demonstrated by knowledge recall and the intellectual skills: comprehending information, organizing ideas, analysing and synthesizing data, applying knowledge, choosing among alternatives in problem-solving and evaluating ideas or actions. This domain on the acquisition and use of knowledge is

predominant in majority of courses. As per the Revised Bloom's taxonomy, the cognitive domain is of two dimensions: Cognitive Processes and Knowledge Categories. The cognitive processes, hierarchically organized, are as follows:

- Remember
- Understand
- Apply
- Analyse
- Evaluate
- Create

These six words, representing the cognitive processes, are also used in common parlance with not so precise meanings. If one wishes to work in the framework of Revised Bloom Taxonomy, these words need to be used as defined by the Revised Bloom taxonomy.

Remembering is retrieving relevant knowledge from long-term memory. The relevant knowledge may be factual, conceptual, procedural or some combination of these. Remembering knowledge is essential for meaningful learning and problem-solving. The two associated cognitive processes are recognizing and recalling. Some action verbs associated with remembering activity include recognize, recall, list, tell, locate, write, find, mention, state, draw, label, define, name, describe, prove a theorem.

Students are said to **understand** when they can construct meaning from instructional messages, including oral, written, graphic communications. These messages are presented to students during lectures, in books or on computer monitors. Students understand when they build connections between the new knowledge to be gained and their prior knowledge. Cognitive subprocesses in this category of Understand and the associated action verbs are as follows:

- Interpret: Translate, paraphrase, represent and clarify
- Exemplify: Illustrate and instantiate
- Classify: Categorize and subsume
- Summarize: Generalize and abstract
- Infer: Find a pattern
- Compare: Contrast, match and map
- Explain: Construct a model

Apply involves using procedures to perform exercises or solve problems. The 'apply' process consists of two cognitive processes: executing—when the task is an exercise (familiar)—and implementing—when the task is an unfamiliar problem. In Executing, a student routinely carries out a procedure when confronted with a familiar task. Implementing occurs when a student selects and uses a procedure to perform an unfamiliar task.

Analyse involves breaking material into its constituent parts and determining how the parts are related to one another and to an overall structure. Analyse includes the cognitive processes of differentiating (determining the relevant and important pieces of a message), organizing (determining the ways in which the

pieces of message are organized) and attributing (determining the underlying purpose of the message). It is considered as an extension of *Understanding* or as a prelude to *Evaluating* and *Creating*.

Evaluate is defined as making judgements based on criteria and standards. The criteria most often used are quality, effectiveness, efficiency and consistency. They may be determined by the student or others. The standards may be quantitative or qualitative. Evaluating includes the cognitive processes of checking (judgements about internal consistency) and critiquing (judgements based on external criteria). However, all the judgements are evaluative.

Create involves putting elements together to form a coherent or a functional whole. Objectives classified as Create have students make a new product by mentally reorganizing some elements or parts into a pattern or structure not clearly present before. Create also refers to objectives calling for production that all students can and will do. If nothing else, in meeting these objectives, many students will create in the sense of producing their own synthesis of information or materials to form a new whole, as in a circuit, a software unit, a mechanism, a structure and so on. The creative process can be thought of as starting the divergent phase in which a variety of possible solutions are considered, as the student attempts to understand the task (generating). This is followed by a convergent phase in which the student devises a solution method and turns it into a plan of action (planning). Finally, the plan is executed as the student constructs the solution (producing).

Categories of Knowledge

While the word 'knowledge' is extensively used by all, there is neither a single agreed definition of 'knowledge' at present nor any prospect of one. Our perspective reflects the idea that knowledge is organized and structured by the learner in line with a rationalist-constructivist tradition and is domain-specific and contextualized. Revised Bloom taxonomy (Anderson et al., 2001) proposes the following four general categories of knowledge applicable to all disciplines:

- Factual Knowledge
- Conceptual Knowledge
- Procedural Knowledge
- Metacognitive Knowledge

Factual knowledge contains the basic elements students must know if they are to be acquainted with the discipline or solve any of the problems in it. The elements are usually symbols associated with some concrete referents or 'strings of symbols' that convey important information. For the most part, Factual Knowledge exists at a relatively low level of abstraction. As our knowledge increases in all fields of inquiry, even experts in these fields have difficulty keeping up with all the new elements. Consequently, some selection for educational purposes is almost always required. The two subtypes of Factual Knowledge are knowledge of terminology and knowledge of specific details and elements.

Conceptual knowledge refers to more general concepts, principles, models, or theories. A concept denotes all the entities, phenomena and/or relations in a given category or class by using definitions. Concepts are abstract, in that they omit the differences of the things in their extension, treating the members of the extension as if they were identical. Classical concepts are universal, in that they apply equally to everything in their extension. Concepts are also the basic elements of propositions, much the same way a word is the basic semantic element of a sentence. Conceptual knowledge includes knowledge of categories and classifications, and the relationships between and among them—more complex, organized knowledge forms. Conceptual knowledge includes schemas, mental models or implicit or explicit theories in different cognitive psychological models. These schemas, models, and theories represent the knowledge an individual has about how a subject matter is organized and structured, how the different parts or bits of information are interconnected and interrelated in a more systematic manner and how these parts function together. Conceptual knowledge includes three subtypes: knowledge of classifications and categories, knowledge of principles and generalization and knowledge of theories, models and structures.

Procedural knowledge is the knowledge of how to do something. The 'something' might range from completing routine exercises to solving novel problems. Procedural knowledge often takes the form of a series or sequence of steps to be followed. It includes knowledge of skills, algorithms, techniques and methods, collectively known as procedures. Procedural knowledge also includes knowledge of the criteria used to determine when to use the procedures. Procedural knowledge is specific or germane to a subject matter or an academic discipline.

Metacognitive knowledge is knowledge about cognition in general as well as awareness of and knowledge about one's cognition. Regardless of their theoretical perspective, researchers generally agree that with development, students will become more aware of their thinking as well as more knowledgeable about cognition in general, and as they act on this awareness, they will learn better (Bransford, Brown, & Cocking, 1999). The labels for this general developmental trend vary from theory to theory but include metacognitive knowledge, metacognitive awareness, self-awareness, self-reflection and self-regulation. Flavell (1979) suggested that metacognition included knowledge of strategy, task and person variables. These are categorized here as follows:

- Strategic knowledge
- Knowledge about cognitive tasks
- Self-knowledge

Affective and Psychomotor Domains

Many researchers firmly believe that non-cognitive factors and skills are equally or even more important than cognitive aspects in educative processes. Grit, tenacity, curiosity, attitudes, self-concept, self-efficacy, anxiety-coping strategies,

motivation, perseverance, confidence are among those frequently referred to as non-cognitive factors. Many of these factors fall into the affective domain.

The Affective Domain is most commonly associated with feelings and emotions. It is usually displayed in the form of positive or negative reactions to given events, objects, behaviours, policies or situations. Affective behaviours are accompanied by varying degrees of feelings and reflect distinct approach or avoidance predispositions. A person's past experience in interacting with the environment shapes the nature and scope of affective responses. (Pierce & Gray, 2013). Pierce–Gray taxonomy classifies the affective domain in terms of increments in cognition.

1. **Perceive** (Emotive Imprinting and Response Setting)
   Sample Outcome: Listen to others' points of view on ethical issues in genetics and biotechnology
   Action verbs: ask, choose, describe, follow, give, hold, identify, locate, name, point to, select, sit, erect, reply and use
2. **React** (Emoting, Recognizing and Controlling)
   Sample Outcomes: Discusses with classmates whether he/she should continue to react in the same way to the ethical issues in genetics and biotechnology
   Action Verbs: answer, assist, aid, comply, conform, discuss, greet, help, label, perform, practice, present, read, recite, report, select, tell and write.
3. **Conform** (Artificial Attitude, Consistent Attitude and Rationalized Attitude)
   Sample Outcomes: Justify the position he/she has taken regarding the use of genetic experimentation from an ethical point of view
   Action verbs: Complete, demonstrate, differentiate, explain, follow, form, initiate, invite, join, justify, propose, read, report, select, share, study and work
4. **Validate** (Examining Values and Accepting Values)
   Sample Outcome: Write a two-page paper explaining why he/she intends to maintain, revise or discard his/her present stance on the current genetic experimentation from an ethical point of view.
   Action verbs: Explain, follow, justify, propose, read, report, select, share, study and work
5. **Affective Judge** (Establishing Value Criteria and Value Judging)
   Sample Outcome: Given multiple stands taken on ethical issues on genetics and biotechnology by different groups, develop criteria based on which one can judge the ethical issues
   Action verbs: Adhere, alter, arrange, combine, develop, complete, defend, explain, formulate, generalize, identify, integrate and modify
6. **Affective Create** (Integrating Values and Inspirational Insight)
   Sample Outcome: Prepare a report that attempts to present his/her ethical standard on genetics and biotechnology supported by the assumptions made and data collected
   Action verbs: Act, create, influence, modify, perform, propose, revise, serve, solve, support.

Learning in the **psychomotor domain** involves motor, muscular activities. Psychomotor learning also occurs in combination with cognitive and affective domain learning. It is demonstrated by physical skills that are acquired through practice. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures or techniques in execution. Riding a bicycle, driving a car, playing a musical instrument, typing, acting and running are dominantly psychomotor activities. The psychomotor activities become important and even dominant in courses of programmes in Theatre, Music, Painting, Sports, Medicine, Nursing, Dentistry, Emergency Medical Services and so on.

Pierce–Gray taxonomy classifies the psychomotor domain in terms of increments in cognition.

1. **Psychomotor Perceive** (Sensory Transmission and Physio Functional Maintenance
   Outcome Sample: Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball.
   Action verbs: Choose, describe, detect, differentiate, distinguish, identify, isolate, relate and select.
2. **Activate** (Physical Outputs, Mimicry and Deliberate Modelling)
   Outcome Sample: Reproduce basic stances, including aramandi, sama, muzhumandi and related exercises.
   Action verbs: Copy, trace, follow, react, reproduce, respond.
3. **Execute** (Task Execution, Operational Execution and Skilled Execution)
   Outcome sample: Perform tha-thai-tham adavus.
   Action verbs: Perform skilfully, react fast, reproduce fast and respond fast.
4. **Manoeuvre** (Inspecting Skills and Selecting Skills)
   Outcome Sample: Link different movements together.
   Action verbs: Adapt, alter, change, rearrange, reorganize, revise, vary, perform, link and prioritize.
5. **Psychomotor Judge** (Establishing Performance Criteria and Performance Judging)
   Outcome Sample: Judge the singing performance of participants.
   Action verbs: Judge, critique, differentiate, mark and select.
6. **Psychomotor Create** (Combining Skills and Performance Insight)
   Outcome Samples: Create a new gymnastic routine.
   Action verbs: Create, develop, perform, arrange, build, combine, compose, construct, design, initiate, make and originate.

Cognitive, affective and psychomotor activities are not independent of one another. Higher levels of affective and psychomotor activities involve more cognitive activities. Instruction needs to pay attention to these dependencies, especially to integrating affective and psychomotor elements into cognitive activities in general courses. For completion, we can also include Spiritual Domain as the fourth domain. Exploration of that domain is beyond the scope of this note. The four domains and their taxonomies are presented as a concept map in Figure 1.

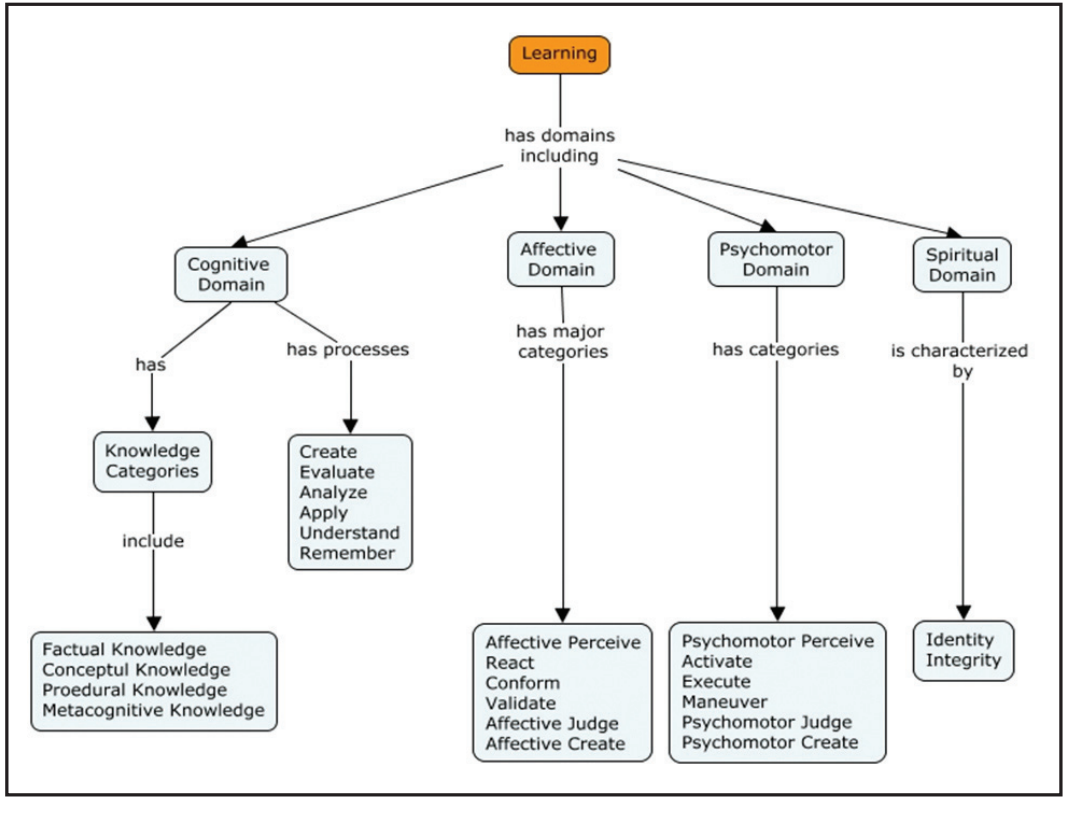

Figure 1. Taxonomy of the Four Domains of Learning

Source: The author.

Taxonomy Table

Cognitive Domain has two dimensions: Cognitive Processes (Levels) and Knowledge Categories. A table with the six rows of cognitive processes and four categories of knowledge can serve as an excellent tool to deal with several issues of teaching and learning, which is known as the Revised Bloom Taxonomy table as shown in Table 1.

Good learning can take place when there is alignment between COs, Assessment Items and Instructional Activities of a course. An element of a course can be tagged by its cognitive level (action verb) and knowledge categories (can be more than one). Alignment among the elements of a course means being in the same cell of the taxonomy table, as shown in Table 2.

Course Outcomes

Graduates of all undergraduate and postgraduate general programmes in India are required to attain the POs identified by the university/college and the PSOs identified by the University or the Department offering the Programme. The POs and

Table 1. Revised Bloom Taxonomy Table of the Cognitive Domain

Cognitive Processes	Knowledge Categories			
	Factual	Conceptual	Procedural	Metacognitive
Remember				
Understand				
Apply				
Analyse				
Evaluate				
Create				

Source: The author.

Table 2. Alignment of the Three Elements in the Revised Bloom Taxonomy Table

Cognitive Processes	Knowledge Categories			
	Factual	Conceptual	Procedural	Metacognitive
Remember				
Understand				
Apply			CO3, IA3, AI3	
Analyse				
Evaluate				
Create				

Source: The author.

PSOs are to be attained through courses, projects and co-curricular and extra-curricular activities in which the performance of the students is evaluated.

Courses are broadly classified into core courses, electives, ability enhancement courses and skill enhancement courses. The POs and PSOs are to be attained through core courses, ability enhancement courses and activities in which all students participate. Courses constitute the dominant part of any programme. Under the present Choice Based Credit System, the courses can be of 3:0:0, 3:0:1, 3:1:0. 4:0:0, 4:0:2, 5:1:0, 0:0:2, 0:0:1, 1:0:2 or 1:0:1 credits. It should be remembered that One Credit is defined as follows:

- One hour of classroom interaction per week over a semester
- One hour of tutorial per week over a semester
- Two hours of laboratory/field work per week over a semester

The course content will have to be realistically adjusted to the number of credits allotted to the course.

The COs are what the student should be able to do at the end of a course. The most important aspect of a CO is that it should be observable and measurable.

We combine the elements proposed by Mager and Anderson and Bloom. The proposed structure of CO statement in cognitive, affective and psychomotor domains, in addition to the common stem, is as follows:

Action: Represents a cognitive/affective/psychomotor activity the learner should perform. An action is indicated by an action verb, occasionally two, representing the concerned cognitive process(s).
Knowledge: Represents the specific knowledge from any one or more of the four knowledge categories.
Condition: Represents the process the learner is expected to follow or the condition under which to act (This is an optional element of CO).
Criterion: Represents the parameters that characterize the acceptability levels of acting (This is an optional element of CO).

Sometimes, it becomes equally essential for a student to perform two cognitive processes on given knowledge elements. Only in such cases, two action verbs are used in a CO statement. It is not an artefact to combine two COs into one. Consider the example of a CO.

Some examples of CO statements are as follows:

Sample 1: Determine the slope from the given topographical map using the Wentworth method.
Action: Determine (Apply)
Knowledge: slope (Conceptual, Procedural)
Condition: given a topographical map, Wentworth method
Criteria: None
Sample 2: Model a spring-mass system as a differential equation
Action: Model (Understand)
Knowledge: spring-mass system (Conceptual)

Condition: as a differential equation
Criteria: None

Sample 3: Understand the paradigm shifts in historical research
Action: Understand (Understand)
Knowledge: paradigm shifts in historical research (Conceptual)
Condition: None
Criteria: None

How many COs should we write for a course? Too small a number does not capture the course in adequate detail and may not serve instruction design that very well. Too many COs make all the processes related to assessment design and computation of the attainment of the COs messy and demanding. A 3:0:0, 3:1:0 and 3:0:1 course should have about six COs. The number of COs of courses carrying a different number of credits can be suitably adjusted.

A CO is to be tagged with the POs and PSOs it addresses, Cognitive Level (R—Remember, U—Understand, Ap—Apply, An—Analyse, E—Evaluate and C—Create), Knowledge Categories (F—Factual, C—Conceptual, P—Procedural and M—Metacognitive), number of Classroom Sessions and/or number of Laboratory/Field Trip hours. Such tagging will facilitate the computation of CO and the PO/PSO attainment. One sample course is presented in the following.

Course: History of Modern India **Credits: 4:0:0**
Programme: BA History

CO	CO Statement	PO/ PSO	CL	KC	Class Hrs
CO1	Describe the role of leaders, nature of events and stages of freedom struggle	PO1, PSO1	R	F	5
CO2	Understand the impact of colonialism on the Indian economy and society	PO1, PSO1	U	C	8
CO3	Explain the concept of nationalism and its phases of growth	PO1, PSO1	U	C	7
CO4	Understand the legacy of the national movement in the making of modern India	PO1, PSO1	U	F, C	6
CO5	Explain the political and social issues in postcolonial India in the context of colonial experiences	PO1, PSO1	U	C	7
CO6	Identify the centres of freedom struggle, the borders during the partition of Bengal and partition of India in the relevant maps and locate states and union territories in India,	PO1, PSO1	R	F, P	4
CO7	Construct arguments in defence of or against historical propositions based on the analysis and evaluation of historical writings	PO1, PSO1	An	C, P	8
	Total Number of Hours				45

Attainment of Outcomes

Designing and conducting undergraduate programmes to attain a set of outcomes is a new experience for Indian Universities. Every programme must attain the POs identified at the Institute/University level and the PSOs identified at the programme level. The POs and PSOs are to be attained through core courses/activities. We need to understand the nature of elements of the POs selected and identify activities that address these elements. It should be remembered that activities planned to address the POs should be amenable to the measurement of their attainment by students.

For example, while everyone considers critical thinking is essential, hardly any programme has activities that directly address and promote critical thinking. Critical thinking is the kind of thinking—about any subject, content or domain—that improves itself through disciplined analysis and assessment. Analysis requires knowledge of the elements of thought; assessment requires knowledge of standards for thought. Understand and solve problems of relevance to society to meet the specified needs using the knowledge, skills and attitudes acquired from humanities/sciences/mathematics/social sciences.

Assuming the COs are written carefully to reflect the intended outcomes of the course and the selected POs and PSOs, the attainment of the COs should lead to the attainment of the POs and PSOs. We can only depend on the common assessment mechanism that exists in the Department offering the programme. The assessment consists of some mid-term tests, end-semester examination, assignments, reports, presentations and laboratory/field performances. If assessments are in alignment with the COs, then their attainment can be readily computed following a non-unique but academically justifiable process. Computing the attainment of POs is a non-unique, imprecise process. However, it is necessary to follow a well-defined process consistently to plan for continuous improvement in the quality of learning.

Misconception

Any framework presented is immediately branded as a straight jacket, and many go impatient about it. If the teacher is a ‘subject expert sage’ and the students are highly competent, there need not be any framework. Such an ideal combination of sages and students is scarce in higher education institutions of India. Even the top-ranked institutions are required to follow the procedures related to positive discrimination and need to be concerned with the issues of equity and access. Outcome-based education merely asks the teacher to communicate at the beginning of the semester, what students are expected to be able to do at the end of a semester rather than the mere list of topics. It has been established through systematic research and field studies, communicating the COs to students at the beginning of the semester makes a significant difference to the performance of students.

No course of any programme can be dealt with in isolation. Each course is an element of a programme and many times related to other courses of the programme. Outcome-based education does not interfere with the academic freedom of the teacher. It merely asks the teacher to follow a process in offering a course. The process consists of writing the COs (what the students should be able to do) in the context of its relationship to the other elements of the programme, design assessment (how to measure the ability of students to do what they are expected to do) and instruction (how the teacher proposes to facilitate the students to acquire the ability to do what they are expected to do). All the decisions in all the three steps of the process are made by the teacher. That is certainly not a definition of a straight jacket. The accreditation agencies require that all institutions write the outcomes, communicate them to stakeholders particularly to students and determine the level of attainment of these outcomes.

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